

It's becoming increasingly difficult for Flash memory to shrink, and therefore, to obey the exponential law. The reason hinges upon the way Flash memory works: memory cells in Flash sticks are transistors whose gates are wrapped in a layer of silicon dioxide that prevents electrons from escaping. The cell holds a one or a zero, depending on the charge in the transistor. The problem is that the silicon dioxide insulating layers are about 90 angstroms thick; and at 80 angstroms, the layer will no longer be as effective.

Although companies such as Intel agree that the issue is currently manageable, making Flash chips with a component size of 45 nm will be difficult. And the 45-nm manufacturing process is set to begin in 2007. Companies, therefore, are looking at alternatives—nanocrystals, Ferroelectric RAM (FeRAM), Magnetic Random Access Memory (MRAM), Ovonics, and polymers.

The use of nanocrystals promises to halve chip sizes by using a lattice of silicon crystals as the insulator, instead of the silicon dioxide layer. At the nanoscale, silicon behaves as an insulator. Silicon-Oxide-Nitride-Oxide-Silicon (SONOS) is another similar alternative that's being researched.

FeRAM is non-volatile, like Flash. However, it's much faster, and consumes less power. The disadvantages include lower density and higher cost. FeRAM stores data by using an electric field to shift the position of individual atoms in crystals.

Ovonics is very different from traditional electron-based storage; it induces a charge by a chemical process. After rapidly heating an alloy substrate similar to the material used in CD-RWs, dots representing ones and zeroes are created in the alloy: a one is encoded by creating a crystalline dot, and a zero is encoded by creating an amorphous dot. The bits are retrieved by measuring the change in electrical resistance between the alloy's crystalline and amorphous states. Ovonics promises to be much faster than Flash, and can be re-written many more times. The problem, currently, is that its fabrication process is expensive.

Intel and other companies are developing polymeric ferroelectric RAM (PFRAM), also known as plastic RAM. The technology involves sandwiching a thin polymer sheet between two perpendicular layers of metal strands. A memory cell is formed at the intersection of these strands, and data is stored by changing the polarisation of the polymer between the strands. PFRAM is a low-power technology. The manufacturing process is simple. And there are cost advantages too: PFRAM requires only a single layer of CMOS circuitry, and can stack up eight layers of inexpensive polymer material. The drawback is speed—read and write speeds don't compare with that of today's Flash chips. Hence, application may be limited to memory sticks.

RAM

Double Data RAM 2 (DDR2) and Graphics DDR3 (GDDR3) are already here, and you can expect to see them in desktop systems very soon. Other new and experimental technologies are on the horizon, all promising that the exponential law for RAM continues for at least a couple of decades to come. It remains to be seen whether RAM will replace the hard disk as mainstream storage.

DDR2

DDR is the current memory standard for most personal computing systems; and DDR2 DIMMs have already been manufactured. DDR2 starts where DDR left off; it's faster, and

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